wireless desktop combo 1000 @ Rs 1,850
- Monitor: You can get a 17-inch CRT from Samsung at Rs 3,500
- Optical: LG GH22NP20 DVD-RAM Drive @ Rs 1,250

Total cost: Rs 43,600 (tentatively). Prices vary, but there are chances you can get a better deal from the vendors in your city if you have time to find out the prices yourself after visiting a few shops.

But let me tell you, you are not bound to this configuration. At a bare minimum, a 500MHz PIII with 256 MB RAM and 80 GB of hard drive space will do, but you will not be able to run many applications. And if you are a hobbyist, you can get away with VMWare server (not VirtualBox or QEMU because of the performance-hit involved).

So what do we have? 2 TB of HDD space, 4 GB RAM, a gaming motherboard (I chose that one because it can be overclocked if need be), a 2.4 GHz QC processor that processes four threads per clock cycle, and a cabinet with a very robust SMPS and plenty of drive bays.

You have a very sleek wireless mouse and keyboard combo, but that's something you can cut down on. Because, at the end of the day, the server will have no monitor and input devices -- everything will be managed via remote X.

Oh, and of course, the UPS system is extra!

The ribcage

To build the basic backbone, or by our analogy, the ribcage of the server, we are going to use (you guessed it!) GNU/Linux. For our distro of choice, we are using a desktop distribution, Kubuntu. You can use Xubuntu or plain and simple Ubuntu as well.

Because this is a very complex procedure, we are going to do this in stages.

Stage 1: Installing the OS

First, get the latest CD of the distro you are going to use. Head to *ubuntu.com*, *kubuntu.com*, or *xubuntu.org* and get a CD. And in case you thought you are going to use the *LFY* December 2008 DVD, it has an installation bug that affects Xubuntu. Get the desktop edition, not the server edition.

Now switch on your PC for the first time, and set the date and time on your BIOS. Search for something like "Boot device priority", and set your PC to try booting from the CD first, and then the hard drive. Save your settings, and exit.

When you reboot, your PC will get stuck and complain about the absence of an operating system. So, insert the Kubuntu CD and hit the reset button on the cabinet.

Directly go to *Install Kubuntu* and start the installation.

The installation process of Kubuntu is just seven screenfulls long. The process is very simple. The only place where you need help is for partitioning.

Create a new MBR on */dev/sda*, and create a 500 MB *ext2* (not 3) for */boot*, a 40 GB XFS for / (root) and a 6 GB swap space. After that, devote the rest of */dev/sda* to a partition mounting under */tmp/ServerTemp*. Use XFS here and then the manual partitioning option.

Complete the installation as usual and reboot into KDE4.

At this point, your server should not be connected or configured for any sort of networking at all.

You will be working as the root user throughout. This is a big security issue, I know, but this is the only hassle-free way of getting everything done perfectly.

Oh, and make a note of disabling all power saving schemes and screensavers. Since this is a server, it may need to stay up for years at a stretch, often without months of desktop activity.

Stage 2: Enable root

Enabling the root isn't a big issue. Open Konsole and type in *sudo passwd root*. Enter a root password twice. Bingo! The root is enabled now.

Stage 3: Internet connectivity

The time has come to connect our server to the Internet. I assume that you are using an 'unlimited' usage broadband connection that has a static IP address. If you don't have a static IP address, ask your ISP for one and write it down somewhere. Having a static IP address is very important.

Now ask your ISP service person to configure your modem to be always-on, i.e., you don't need to enter a password to access the Internet. This constitutes entering the user name and password into the modem itself so that it authenticates every time it switches on.

The Star Bus Topology

A topology is the arrangement in which computers on a LAN are physically connected together. There are many such topologies, such as Ring, Star, Bus, Mesh, etc. Since the Ethernet is not connection-centric, any sort of arrangement works, but these topologies have a wide use, as they are the only way to keep the physical connections organised.

The Star-Bus topology is a hybrid topology, using the best of both the Star and Bus worlds. In this arrangement, a single high-capacity line, called the Bus, runs from the server to a hub. From the hub, many more lines run to one client each, or to other hubs. In the latter case, the hub-to-hub line becomes another bus. In a schematic diagram, each of these clients on a hub is represented by spokes running from the hub to the client; thus the Star.

In a Star-Bus topology, you have flexibility as well, because you can have different servers at different levels, each tending to the levels below it.